

EQUIPPING NEXT-GEN EV CHARGING WITH all-weather heating and cooling

Shell, as part of Powering Progress, targets installing more than 500,000 electric-vehicle charge points by 2025. Future charging solutions will address current challenges including long recharging time, low charging speed in cold temperatures and the risk of battery thermal runaway during fast charging. Since 2018, Shell and Tsinghua University, China (see boxed text, [Shell and Tsinghua University joint research partnership](#)), have been collaborating on a long-term co-engineering technical project to develop a novel high-performance charging (HPC) system. The charging system will have 350 kW of power and will control a patented bidirectional pulse-heating function for heating cold batteries and an external cooling system for controlling battery temperature during supercharging.

Introduction

Shell, Tsinghua University and other partners demonstrated the charger and a pilot charging station during the Beijing 2022 Winter Olympic Games and successfully proved the feasibility of the technologies, including bringing batteries on a common passenger electric vehicle (EV) from 0 to 25°C at 6.7°C/min and recharging to 100-km range in 5 min without overheating. The pilot charging station forms an intelligent microgrid by implementing solar panels, energy storage batteries and heavy-duty vehicle battery swapping, thereby demonstrating a possible low-carbon scenario for e-mobility integration. In the future, bidirectional pulse heating and external thermal management will be further evaluated before they enter the market.

Background of e-mobility development in China

Based on data from the Ministry of Industry and Information Technology, [China has had the world's largest automotive market for 13 years](#). And, in recent years, motivated by government support and sustainable initiatives such as the 2060 net-zero ambition, the Chinese EV market, including battery electric vehicles (BEV), plug-in hybrid

vehicles (PHEV) and range extender electric vehicles (REEV), is developing fast. China is considered a [leader in e-mobility given the fast growth of both its EV market and charging infrastructure](#), and strong innovation in the EV and batteries areas.

EV and charging infrastructure market

According to data from the China Association of Automobile Manufacturers, China's new energy vehicle (NEV, meaning BEV, PHEV, REEV and fuel cell vehicles) sales from Q1 to Q3 2022 reached 4.5 million [\[Ref 1\]](#) and have already surpassed the total NEV sales volume of 3.3 million in 2021 [\[Ref 2\]](#). This also means that the market share of NEVs has now increased to 27.1%, from 5.8% in 2020 [\[Ref 2\]](#). The China National Development and Reform Commission stated that NEVs in operation, or the vehicle population, in China was [7.84 million in 2021](#), representing 2.6% of the total vehicle volume and about a 50% share of the global NEV market. The [China Society of Automotive Engineers](#) estimates that the NEV market share will grow to 50% in 2035, and BEVs will be strongly promoted as private and business cars, taxi and rental fleets, and short-trip commercial vehicles. By benefiting from long-term research and development, as well as significant policy support and subsidies from the government, China has the largest EV industry in the world; with this tremendous EV market, the industry is becoming market-oriented and has incubated many successful enterprises, including vehicle manufacturers like BYD and battery manufacturers such as CATL.

Compared with the size of the EV market, charging infrastructure development is not yet sufficient but is starting to grow faster. Statistics from the China Electric Vehicle Charging Infrastructure Promotion Alliance show that there were [4.448 million charging points in mainland China by September 2022](#); of these, 1.636 million are public. Direct current (DC) charging is used in 0.704 million chargers and the rest use alternating current (AC). Meanwhile, the ratio of

Shell and Tsinghua University joint research partnership

Tsinghua University is in Beijing, China. It is a national public research institution, well known for its strong research capability in science and engineering. Tsinghua University is not only a prestigious academy but also an important idea bank for China's national government, hence its impact on policy design and industry development strategy is powerful. Shell and Tsinghua established the Tsinghua (School of Vehicle and Mobility)–Shell Joint Research Centre for Clean Mobility in 2017. Research focuses primarily on developing solutions for the automotive sector, including the development of lubricants, low-carbon fuels, charging technology, battery thermal management, green hydrogen and more. The partnership also extends to cohosting impactful events such as the Shell Helix product launch, Shell Powering Progress summit, China, and the joint technology forum.

Approximate power rating, kW		30	60	90	120	160	180	240	350
Charger manufacturer	Starcharge	✓	✓		✓		✓		✓ (360 kW)
	TELD	✓ (42 kW)			✓			✓ (240/250 kW)	✓
	iCharge	✓	✓					✓	✓ (360 kW)
Vehicle manufacturer	NIO				✓ (105 kW)		✓		
	Xpeng			✓			✓		
	Tesla				✓ (V2)			✓ (V3, 250 kW)	✓ (V4)

High voltage platform: 500, 750 and 1,000 V

new chargers to new EVs is about 1:2, which means the rate of growth of charging infrastructure is catching up with that of EVs. Given the future demand for high-performance charging (HPC) facilities, the infrastructure needs developing to be direct-current, high-power, intelligent and interactive.

Trend to DC fast charging development

Compared with AC charging, DC charging provides power directly to the battery and allows much faster charging speeds since it bypasses the limitations of the onboard charger and required conversion. Hence DC fast charging is growing rapidly. The data in Table 1 show the key charging solution providers and equipment manufacturers. Almost all companies in China, except Tesla and Geely Zeekr, select high-voltage rather than high-current charging to enable high power, mainly because high-current charging generates more heat and requires a more complex thermal management system for the EV and the charger. Recently, the average output power of new DC chargers has increased to > 120 kW; however, HPC > 250 kW may not prevail in the short term, as updating the charger and vehicle to ultrahigh voltage will present many challenges, such as redesigning the electrical system and the high cost of hardware.

The challenges to solve

As EV charging power is growing momentarily, the restrictions on recharging lithium-ion batteries, the main form of battery in EVs, are becoming concerning. For both consumers and manufacturers, solving the issue of long charging time, especially in cold environments, and the safety concerns relating to battery thermal runaway are most demanding.

Long charging time

The charging rate is usually designated by C, defined as the charge current divided by the battery nominal rated capacity. A higher C means a higher charging rate; for example, if a battery can be charged at 1C it will charge from 0 to 100% in 1 h, whereas at 5C the full charging time is only 12 min. The charging rate is typically low compared with refuelling a conventional vehicle powered by

an internal combustion engine because fast charging generates more waste heat and accelerates battery degradation, which can lead to fire and even battery explosion. Hence, the battery management system (BMS) sets a charging rate limit to mitigate such risks [Ref 4]. As a result, most EVs take a relatively long time to be fully recharged; as shown in Figure 1, it takes about 1 h for a 50-kW DC charger to bring a BEV state of charge (SOC) to 80%.

Increasing the charging rate is critical to reducing range anxiety among consumers. The rate depends on the chemical nature of the battery pack and the BMS control strategy, and the proper charging equipment is necessary to provide the power required. Potentially, with a 350-kW high-performance DC charger and a high C-rate capable battery pack, the total charging procedure can be reduced to just a few minutes, and an approximate 200-km range can be achieved in 10 min or less.

Charging difficulty in cold environments

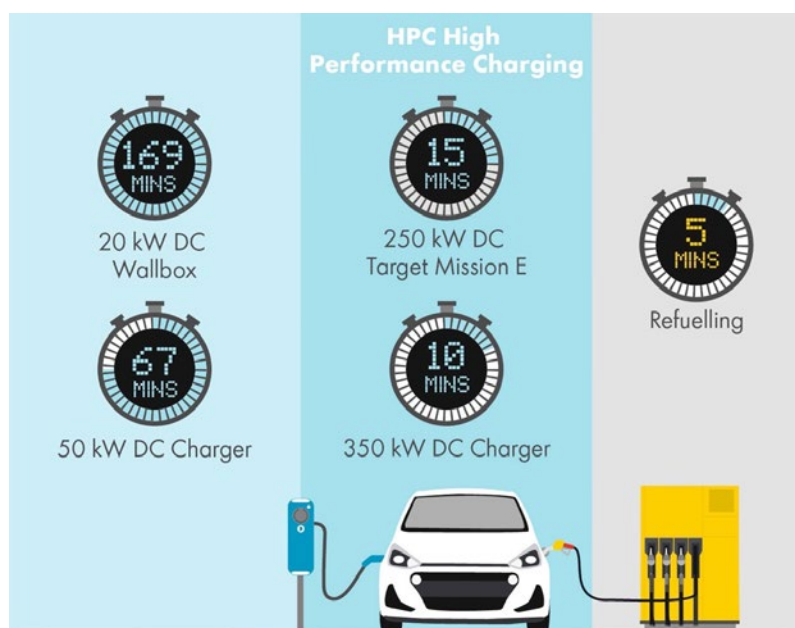
The charging rate can be even slower if the ambient temperature is low. In cold environments, lithium ions in the electrodes and the

TABLE 1

Charging power of products from representative charger and vehicle manufacturers in China [Ref 3].

FIGURE 1

Typical charging times for 80% battery capacity lift.



Venting

- Cell heats up
- Electrolyte is degassing
- No fire if the internal or external heat stops

**Thermal runaway**

- If heat is still increasing, cell internal materials react with each other
- Cell starts burning explosively

**Propagation**

- Fire in a single cell "infects" other cells
- Chain reaction
- Dangerous increase of heat, pressure and gases

**FIGURE 2**

Illustration of the pathway to battery thermal runaway.

electrolyte diffuse slowly, and the kinetics of intercalation can be sluggish, which lowers the charging rate. If the BMS forces the charging rate to be fast in such conditions, the risk of lithium plating during charging greatly increases and impacts battery capacity retention. Meanwhile, the efficiency of charging equipment is not ideal in low temperatures; Trentadue [Ref 5] reported that the power conversion efficiency of a 50-kW charger is only 39% at -25°C compared with 93% at 25°C . Thus, the battery should be preheated before normal charging starts.

A positive temperature coefficient (PTC) heater is the most widely used equipment for battery preheating, but it is bulky and slow. Hot fluid heating methods using a liquid or a heat pump are proven to be more effective, but they may occupy more space inside the vehicle [Ref 6]. At extremely low temperatures ($< -10^{\circ}\text{C}$), the heating rates of both PTC and hot fluid heating are usually below $1^{\circ}\text{C}/\text{min}$, which is not ideal [Ref 7, 8].

Thermal runaway risk

As illustrated in Figure 2, if the heat generated by the battery cannot be dissipated in time, it could lead to dire consequences such as fire and explosion. This process is called thermal runaway and it is an uncontrollable reaction. To prevent it, the battery pack needs a reliable thermal management system to remove heat and keep the battery temperature within an acceptable and predictable range. Most OEMs adopt an indirect-cooling system that removes heat from the battery pack with a water-glycol-based coolant thanks to its ability to maintain a uniform temperature distribution, favourable specific heat capacity and

good thermal control. In an indirect-cooling system, coolant flows inside cooling pipes or plates that are in contact with the battery cells and carry the rejected heat to a chiller. The heat is dissipated to the environment and the coolant returned to repeat this cycle. Recently, the indirect-cooling system alone is facing more challenges to support ultra-fast charging, as the thermal resistance between the cooling channel and the battery is high and efficiency is hard to improve further, and it is hard to improve the heat-exchange capacity of a chiller in a confined space.

The Shell solution: External thermal management

The battery itself needs to be improved to address these challenges. In addition, enhanced charging equipment could improve the charging process for existing vehicles and add extra value to future batteries and vehicles, especially through battery heating before charging and battery cooling during charging. To gain a strong position in the market and accomplish the target of installing 500,000 charge points by 2025, Shell is investing in charging technology innovation globally.

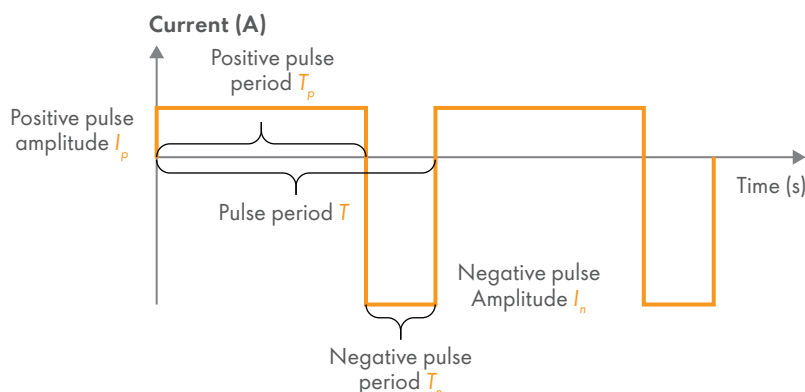
Shell and Tsinghua University set up the Joint Research Centre for Clean Mobility in 2017, and in 2018 they started a project to evaluate two innovative techniques to improve both low-temperature charging and temperature control of battery cells: bidirectional pulse heating and external cooling – together, known as external thermal management.

Bidirectional pulse heating

Before the battery reaches the optimal charging temperature of $25\text{--}45^{\circ}\text{C}$ [Ref 9], a preheating process is needed to reduce degradation and safety risks. Bidirectional pulse heating is an internal heating method that uses a periodic charging and discharging current to generate Ohmic (also referred to as Joule or resistive) and reaction heat inside the battery. Compared with external heating methods such as PTC heating, bidirectional pulse heating is highly efficient with a high heating rate and good temperature uniformity [Ref 10]. An external pulse control device is needed to control pulse voltage, current and time. A typical pulse heating current is shown in Figure 3.

FIGURE 3

Bidirectional pulse heating method.



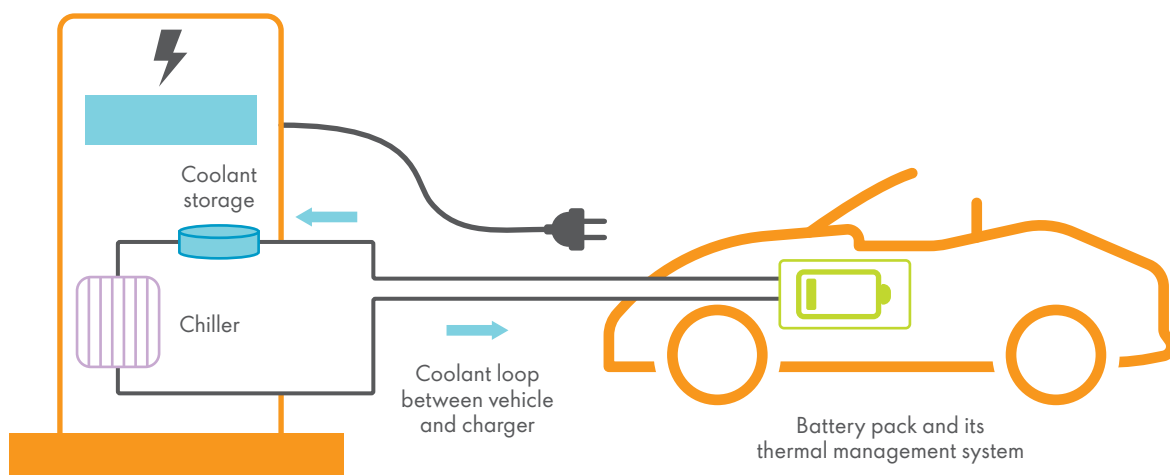


FIGURE 4
Example internal
cooling schematic.

External cooling

Existing cooling systems for batteries inside a typical vehicle cannot control cell temperature in an ideal range for a long time during supercharging (> 300 kW) because they are restricted by space, cost and the efficiency plateau of indirect liquid cooling [Ref 11]. External cooling uses an extra cooling medium to better remove heat from the battery and then circulates the cooling medium to an external and more efficient heat exchanger to dissipate more heat than an internal vehicle cooling system can. Figure 4 shows the proposed change regarding coolant circulation from internal to external cooling.

Between 2018 and 2020, Shell and Tsinghua University validated these innovative methods in a laboratory at module level, and the outcomes were promising. The experimental results from the pulse heating tests showed only 1% battery capacity decay after 170 h of continuous heating at 11°C/min at battery cell level; and the external cooling tests with a three-sided cooling plate arrangement have shown the maximum temperature of battery cells

remains below 50°C even when the charging speed is 5C. As summarised in Figure 5, data from the simulation show that, if the battery pack temperature is -10°C, by applying pulse heating and external cooling a maximum charging speed of 3.6C can be achieved, and the total charging time can decrease from 110 to 21.8 min. Based on the data the team obtained, Shell and Tsinghua University patented the pulse heating development algorithm.

As a result of this validation, Shell and Tsinghua University decided to follow up by adapting these techniques to produce a prototype charging system that can control the temperature of vehicle batteries externally. The charging system prototype has three components: a power cabinet, a charger and a cooling cabinet (Figure 6).

Power cabinet

Bidirectional DC-DC modules are installed inside the power cabinet to convert DC from the energy storage system (ESS) to DC for the vehicle, and to convert DC from the vehicle to DC for the ESS when pulse heating is activated. >>>

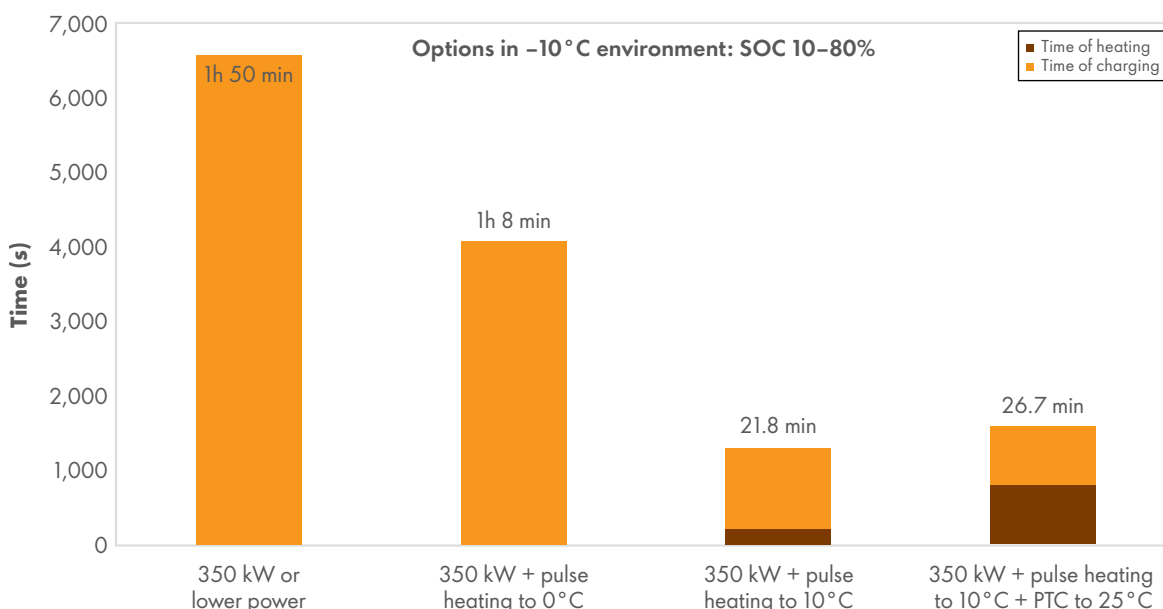


FIGURE 5
Influence of
external thermal
management on
charging time.



FIGURE 6
External thermal management charger prototype, left to right: power cabinet, charger and cooling cabinet.

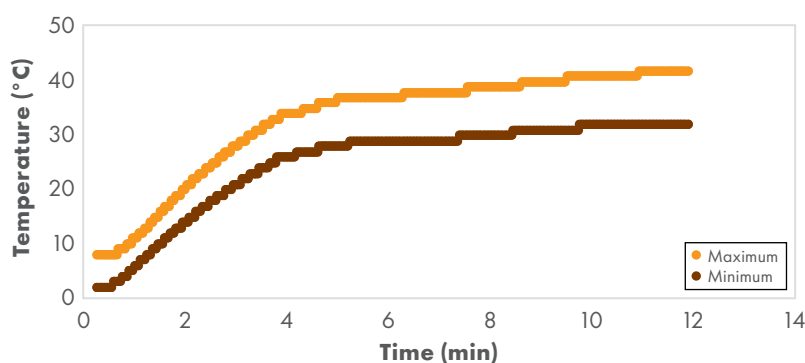


FIGURE 7
Temperature data during vehicle charging experiment.

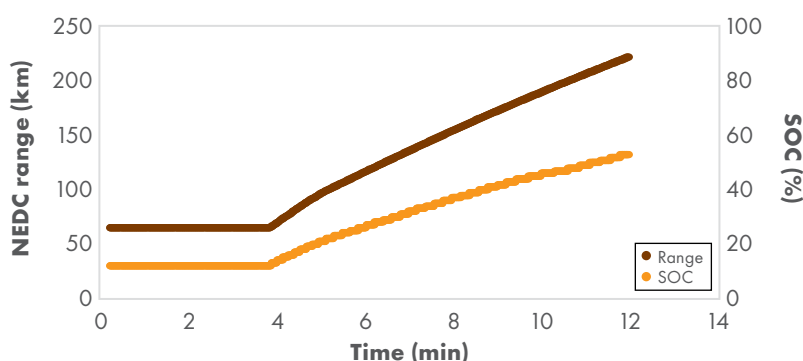
Charger

The charger has a screen displaying the user interface to manage the charging process. The charger is compliant with Chinese GB/T standards, and one of the two connectors is manufactured based on the Chaoji standard, which allows a maximum power of 900 kW, for potential future upgrade.

Cooling cabinet

The cooling cabinet stores extra coolant outside the vehicle. When the charger and vehicle are connected, the pump inside the cabinet starts a

FIGURE 8
Range and SOC data during the vehicle charging experiment.



circulation between them. The coolant is injected into the cooling plates of the EV battery pack and excess heat is removed through a large chiller inside the cabinet. More heat can be removed this way than the vehicle's thermal management system can manage, so the batteries are cooled more effectively.

The coolant used is Shell Rotella ELC Nitrite Free Pre-Diluted 50/50 antifreeze/coolant. This is an ethylene glycol-based coolant that contains the proven nitrite-free Shell Rotella ELC organic acid technology for exceptional long service life, outstanding heat transfer and corrosion protection in all weather conditions.

With this external thermal management setup, this charging system can achieve the following:

- Before charging starts, the battery cells of the EV can achieve an average heating rate of 5°C/min in low-temperature environments until the cells reach the optimal charging temperature (about 25°C).
- The vehicle can be recharged to a range of 100 km or even further within 5 min using the advanced heating and cooling methodology. Range can also be influenced by other factors such as EV battery chemistry and thermal management system, specific vehicle performance, calibration cycle used and natural temperature variations.
- During charging, the maximum temperature of the vehicle battery pack can be controlled within a safe range, no higher than 45°C.

Progress of technology development

The external thermal management methodology and the corresponding charging technology, jointly developed by Shell and Tsinghua University, still need further assessment before they can be introduced to the market. Nevertheless, an important milestone has been reached. The team have successfully demonstrated high-performance, all-weather charging of a common electric passenger vehicle made by a Chinese manufacturer. The concepts of fast charging, pulse heating and external cooling have been evaluated and proven at the vehicle level.

In the experiment, the ambient temperature was < 0°C, the initial minimum temperature of the battery pack was 0°C and the battery SOC was low, at 12%, before the charging process began. First, pulse heating was activated. The data in Figure 7 show that the pulse heating stage lasted 3.7 min; during this period, the lowest battery temperature observed rose from 0 to 25°C, and the highest temperature increased from 8 to 33°C. The highest heating rate recorded was 7.2°C/min at 2.45 min, after which the heating rate fell. The average heating rate was 6.7°C/min.

Fast charging started immediately after pulse heating had achieved the optimal charging temperature of 25°C. Restricted by the vehicle BMS, the maximum charging power is 250 kW, which is lower than the maximum output power of the charger (350 kW). Charging power of 250 kW could be maintained for the first 30 s, then the power decreased to about 235 kW for 35 s and then charged at an average rate of 150 kW for the remainder of the session. During the charging process, the vehicle SOC increased from 12 to 51% in 463 s (7.7 min) while the range, calibrated based on the New European Driving Cycle (NEDC), increased from 65 to 215 km (Figure 8). Hence, the vehicle was successfully recharged to a 150-km range in 8 min, for this vehicle with a 73.5-kWh battery and an energy consumption of 14.6 kWh/100 km. The battery temperature was maintained below 45°C through the 150-km recharge cycle with the help of external cooling.

Demonstration of technology leadership

To gain better exposure for the technology, the project team rented a plot in Zhangjiakou, China, 190 km away from Beijing and a co-host city of the 2022 Winter Olympic Games, and constructed a demonstration station including the Shell external thermal management charging system. In addition to the charging system, this demonstration station has a 120-kW DC charger, a 15-kW photovoltaic system, an ESS and screens to display charging data and the energy flow of the station (Figure 9).

Figure 10 shows the 3D layout of the supercharging demonstration station. A battery swapping station, located near the Shell Recharge station, is owned by one of Shell's partners and is designed for battery replacement for electric heavy-duty vehicles. Together, these two stations make the whole station the first integrated e-mobility service demonstration with the features of low-temperature fast charging, heavy-duty



FIGURE 9
The Shell charging demonstration station in Zhangjiakou.

vehicle battery swapping, photovoltaic electric generation and an ESS.

In January 2022, the Shell Fuels and Lubricants Technology team and Shell China Mobility Marketing held an opening ceremony for the supercharging demonstration station. The Shell China Country Chair, Vice President Fuels and Lubricants Technology and Vice President Mobility China were invited to this event. All partners, including Tsinghua University and BAIC, participated in the ceremony and viewed a demonstration of the Shell charging station. Several publications have been issued based on the demonstration station and shared externally, mainly in China.

Scope of future work

Shell will continue to explore the HPC market. In May 2022, Shell China officially announced the opening of an ultrafast charging point having >>>

FIGURE 10
Layout of the supercharging demonstration station.



a 360-kW power capacity in Wuhan. This is the fastest charging capacity for passenger EVs in mainland China so far. This 360-kW charger is a simplified version of the one demonstrated during the 2022 Winter Olympics in Zhangjiakou, without the pulse heating and external thermal management function.

Shell, Tsinghua University and some vehicle manufacturers have agreed to further validate bidirectional pulse heating technology before it can be commercialised. The target is to further increase the battery preheating rate with higher heating power while testing other effects such as power loss and battery degradation. Shell will also seek opportunities to work with other partners on promoting bidirectional heating in China and around the world.

The design of the external cooling system can be further improved, and the concept may be transplanted to other applications such as ESS.

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Review:

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